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ATTENTION: Mr. W. Wolensky December 3, 1957

SUBJECT: Memory Addressing Scheme

On Monday, December 2, 1957, Dr. G. A. Blaauw described the following addressing scheme, which has been agreed upon by Stretch Planning and Product Planning, for the Stretch system.

Addresses 0 through 13 or 0 through 15 are assigned to special registers in the CPU.

Up to 48 words of high speed register memory will be available in units of one word. They will be assigned addresses from 15 through 63. Those to be used as index registers will be assigned addresses 49 through 63. A high speed, .5 usec core memory will be available in boxes of 1024 words. Either 0, 1, 2 or 4 boxes may be used in a system. They will be assigned a continuous set of addresses starting at zero. Where an address refers to both a register and a word of core memory the word of core memory will not be available. Interleaving of .5 usec memory will be made available by using the low order bit of the address as a frame address when 2 boxes are used and using the two low order bits as a frame address when 4 boxes are used.

Two (2) usec memory will be available in boxes of 16, 384 words. Either 0, 1, 2, 4, 8, 12 or 16 boxes may be used in a system. Two (2) usec memory will be assigned a continuous set of address starting with an address one higher than the highest address in .5 usec memory. Before access is made to a 2 usec memory, the address will be modified to one of a continuous set starting at zero as proposed in Basic Computer Memo #3 by S. G. Tucker.

Interleaving will be provided for 2 usec memory by using the low order bit of the modified address as the frame address if there are 2 boxes. If there are 4, 8, 12 or 16 2 usec boxes two low order bits and as many high order bits as required will be used as a frame address. This will provide interleaving in blocks of four boxes.

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Number of 2 usec Boxes	Assignment of Bits in Modified Adders
0	----------
1	XXXXAAAAAAAAAAAAAA
2	XXXAAAAAAAAAAAAAAF
4	XXAAAAAAAAAAAAAAFF
8	XFAAAAAAAAAAAAAAFF
12	FFAAAAAAAAAAAAAAFF
16	FFAAAAAAAAAAAAAAFF

X = Not Used. F = Frame Adr. Bit A = Array Adr. Bit

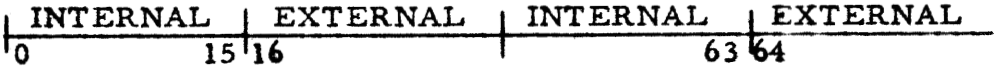
Invalid addresses may be detected by the presence of a 1 in an X bit or, in the case of 12 boxes, by the presence of a 1 in both high order F bits.

This scheme provides a set of addresses with no holes, and a loss in core memory equal to the number of high speed register words used plus 15.

It is felt that the following items need further clarification or definition.

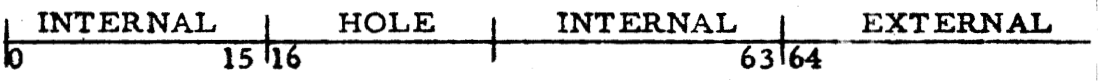
I. Some form of control panel will be required to alter the address modification and the invalid address detection procedures to correspond to the memory complement in use. Should this be a switch panel, a plug board, a wiring change or something else?

II. In the system just described, addresses 0-13 address internal registers. It would be much simpler to identify these if it were extended to 0-15 as has been suggested. If less than 48 High Speed registers were used, the assignment would be as follows:

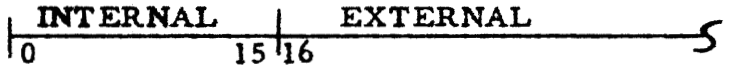


An address greater than 63 could immediately be classified as external and the proper access would then be started. An address less than 63 would have to be decoded in order to determine whether it referred to an internal register or a core memory location. Access time to core memory addresses less than 64 would be increased accordingly.

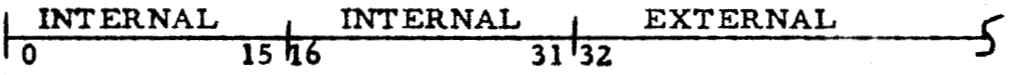
The circuit might be simplified by not allowing core memory address less than 64. This would result in holes where no high speed registers existed. The assignment would then be as follows:



Another approach would be to permit only 0, 16, or 48 high speed registers. This would result in a simple determination of internal and external memory locations. The assignment would then be as follows:



or



or



The logical design in progress will be aided by definite answers to the above questions.

S. G. Tucker

SGT/emb

Mr. W. Wolensky December 3, 1957

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